FADI KELADA

Planning-tools consultant at Enedis | Ph.D. in Electrical Engineering

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EXPERIENCE

Consultant, Technical Department (planning tools)

Enedis

Nov 2024 – present

Expert on the low-voltage network planning tools. Pilot of the project to redesign the planning calculation tools for 2032 (Projet refonte des outils de calcul).

Doctoral researcher

Grenoble Electrical Engineering Lab (G2ELab), Université Grenoble Alpes

Oct 2021 – Sep 2024 · Grenoble, France

Small-signal stability and EMT modelling of low-inertia grids with high shares of inverter-based resources; grid-forming converter control. Author of G2ELin. 100+ teaching hours, 2 interns supervised.

R&D Engineer (internship)

Supergrid Institute

Mar 2021 – Aug 2021 · Lyon, France

Developing power-system electromagnetic transient (EMT) models and simulations in OpenModelica for reference grids incorporating HVDC and hybrid AC-DC networks.

Intern on Energy Management

Schneider Electric

Feb 2020 – Aug 2020 · Cairo, Egypt

Graduation project internship: led my team to develop new energy management solutions for air-conditioning systems. The project won first prize.

EDUCATION

Ph.D. in Electrical Engineering

Université Grenoble Alpes, Grenoble Electrical Engineering Lab (G2ELab)

Oct 2021 – Sep 2024 · Grenoble, France

Thesis: “Study of dynamics and stability in low inertia power systems with high penetration of inverter-based resources”, defended on 24 September 2024. Supervisors: Prof. Nouredine Hadjsaid and Dr. Jérôme Buire.

M.Sc. in Electrical Engineering

Grenoble INP, Université Grenoble Alpes

Sep 2020 – Sep 2021 · Grenoble, France

Valedictorian. Grade: 16.48/20.

ABOUT ME

Power-systems engineer at Enedis, where I work on the network planning tools, with a PhD on the dynamics and stability of low-inertia grids dominated by inverter-based resources. I build open tools for stability analysis (G2ELin) and for teaching power systems (Omega Lab).

KEY FIGURES

2 Journal articles
4 Conference papers
57 Citations (h-index 3)
1 Open-source tool
100+ Teaching hours
2 Interns supervised

ACHIEVEMENTS

IEEE SYP R8 Congress Secretary
Co-chaired the IEEE R8 SYP congress held in Grenoble in 2024.

Best Energy Management Solution
Led my graduation project team to win first prize in Schneider Electric’s undergraduate projects competition.

Scholarship Holder
Ranked first in my Bachelor’s class; received a fully funded French government scholarship for my Master’s and PhD.

SKILLS

Power systems
Small-signal stability · EMT simulation · Grid-forming & grid-following control · Modal analysis · Network planning

Simulation tools
MATLAB · Simulink · OpenModelica · EMTP-RV · Hypersim · DIgSILENT PowerFactory · pandapower · HOMER

Programming
Python (NumPy, SciPy, SymPy, FastAPI, pytest) · C/C++ · Modelica · TypeScript

B.Sc. in Energy & Renewable Energy Engineering

Ain Shams University

Sep 2015 – Aug 2020 Cairo, Egypt

Valedictorian. GPA: 4.0/4.0.

PUBLICATIONS

Journal articles

- Fadi Kelada, Jérôme Buire, Nouredine Hadjsaid. "Revisiting Small-Signal Modeling for Analyzing Fast Dynamic Interactions in Converter-Dominated Power Systems". *IEEE Transactions on Power Systems, 40(5), 3819–3832*, 2025. doi:10.1109/TPWRS.2025.3550287
- Jérôme Buire, Fadi Kelada, J. Marchand, Nouredine Hadjsaid. "Tuning Grid-forming Converters for Optimal Allocation of Virtual Inertia and Damping in Low-Inertia Power Systems". *International Journal of Electrical Power & Energy Systems, 178, 111977*, 2026. doi:10.1016/j.ijepes.2026.111977

Conference papers

- Fadi Kelada, Jérôme Buire, Nouredine Hadjsaid. "Stability of Distribution Networks under High Penetration of Inverter-Based Resources: Highlight of Novel Interactions". *28th International Conference on Electricity Distribution (CIRED 2025), IET Conference Proceedings 2025(14), 2919–2923*, 2025. doi:10.1049/icp.2025.2210
- Fadi Kelada, Jérôme Buire, Nouredine Hadjsaid. "Comparative Study of Decentralized Grid-forming Converter Controls for Inverter-based Microgrids". *2023 IEEE Power & Energy Society General Meeting (PESGM), Orlando, FL*, 2023. doi:10.1109/PESGM52003.2023.10252590
- Fadi Kelada, Jérôme Buire, Nouredine Hadjsaid. "Placement of Virtual Inertia in Islanded Distribution Networks with High Penetration of Inverter-Based Resources". *27th International Conference on Electricity Distribution (CIRED 2023), Rome*, 2023.
- Fadi Kelada, Jérôme Buire. "Contribution of Power Converters in Frequency Stability of Low-inertia Power Systems". *Conférence des Jeunes Chercheurs en Génie Electrique (JCGE), Le Croisic, France*, 2022.

Thesis

- Fadi Kelada. "Study of Dynamics and Stability in Low Inertia Power Systems with High Penetration of Inverter-Based Resources". *Ph.D. thesis, Université Grenoble Alpes*, 2024. https://theses.fr/2024GRALT065

Software

- Fadi Kelada. "G2ELin: An Open-Access Power System Linearization and EMT Simulation Tool". *github.com/FKELADA/G2ELin*, 2023.

Web & tooling

React · Svelte · Next.js · Git & GitHub Actions · Docker · LaTeX

LANGUAGES

Arabic
English
French
German



PROJECTS

G2ELin

Open-access power-system linearization and EMT simulation tool: MATLAB/Simulink library and a Python port with a FastAPI backend and web interface, 530+ automated tests and MATLAB parity checks.

Omega Lab

Interactive course on electrical power systems, from RLC circuits to grid stability: 67 lessons with live simulations, in French and English.

VOLUNTEERING

Catechism servant

Coptic Orthodox Church of St Michael & St George, Villejuif

Serve in the catechism service, and built Khedma, the servants' web app: classes, attendance, birthdays and weekly follow-up calls, in French and Arabic.

IEEE Region 8 Student and Young Professionals (SYP)

IEEE

12 months of volunteering; co-chaired the IEEE R8 SYP Congress held in Grenoble in 2024.